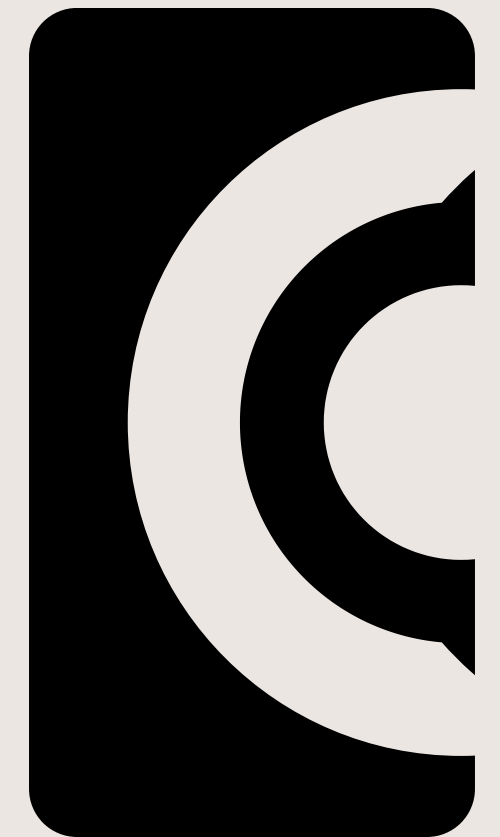


Iteration 2

EasySpot

Parking Analytics Platform



agenda

Architecture Notebook

Prototypes for the core user
stories

Key quality requirements

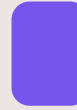
The primary focus of EasySpot is to enable a system that has a low latency data flow for real-time occupancy events and a persistent and scalable database that stores historical and statistical data.

Non-functional requirements and constraints

 **Performance and Responsiveness**

 **Scalability**

 **Data Consistency and Integrity**

 **Deployment and Portability (Constraints)**

Key quality requirements

The primary focus of EasySpot is to enable a system that has a low latency data flow for real-time occupancy events and a persistent and scalable database that stores historical and statistical data.

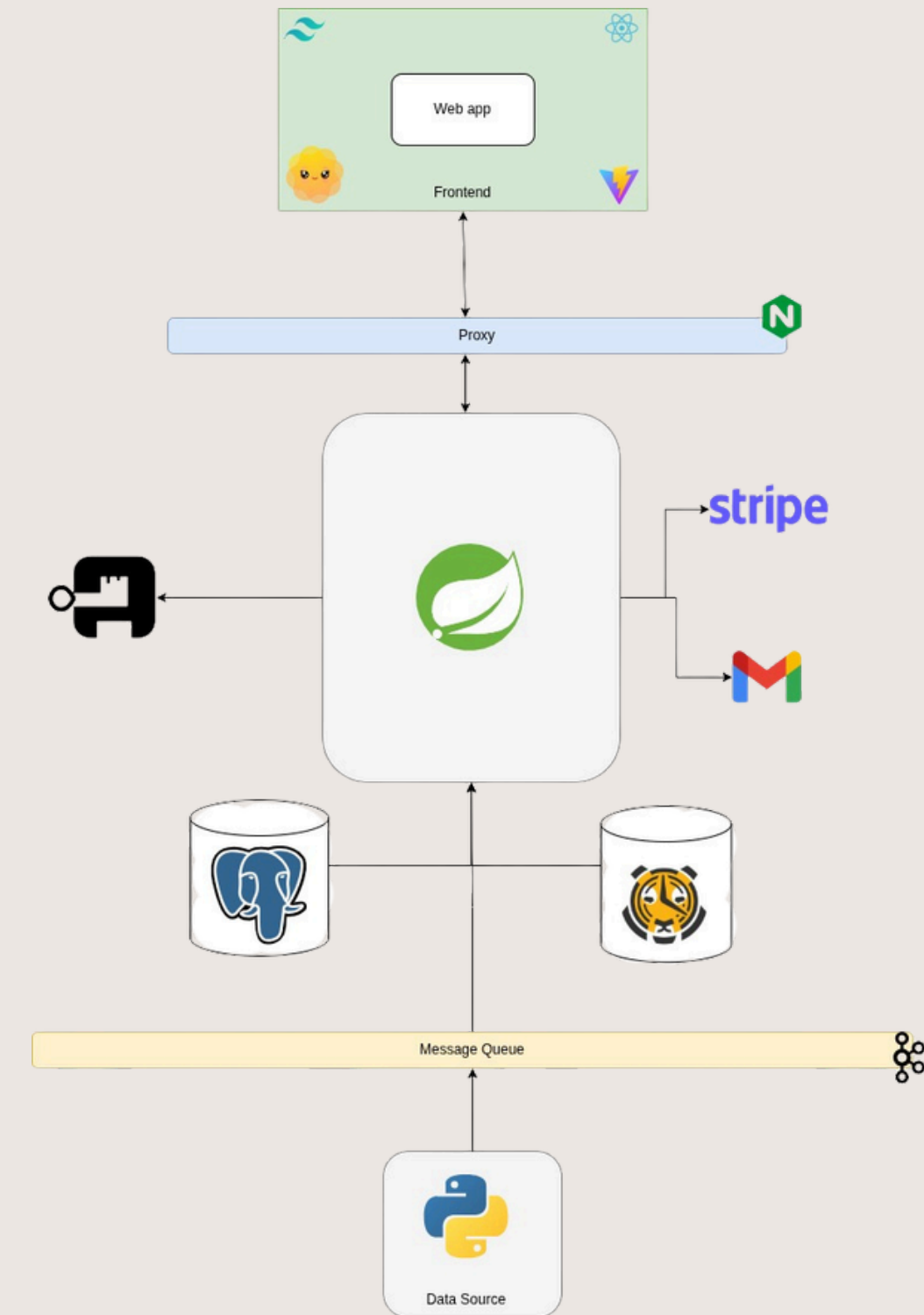
Functional requirements

-  Real-time Monitoring & Alerts
-  Parking Simulation

Layered Architecture

System components

EasySpot is structured into four main components: Data Sources, Message Queue, Backend, and Frontend. These components work together to ensure a smooth flow of data from generation to user interaction, enabling real-time processing and efficient system performance.



Data Flow & User

Interaction

Data & Processing

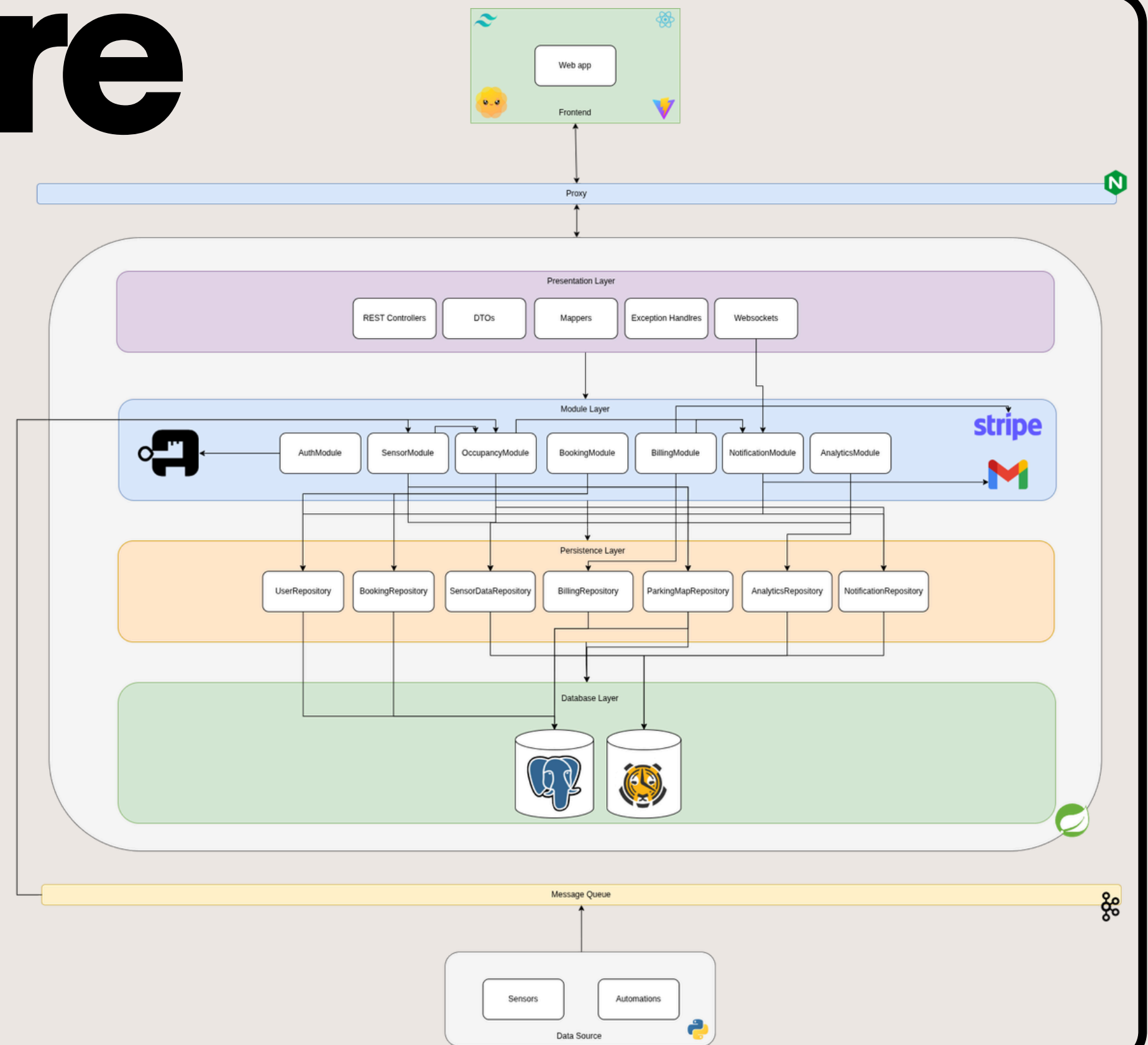
The system uses simulated sensors built in Python to generate real-time data. This data is streamed through Apache Kafka and processed by the backend using Spring Boot. The system stores information in PostgreSQL for relational data and TimescaleDB for time-series data.

User Side

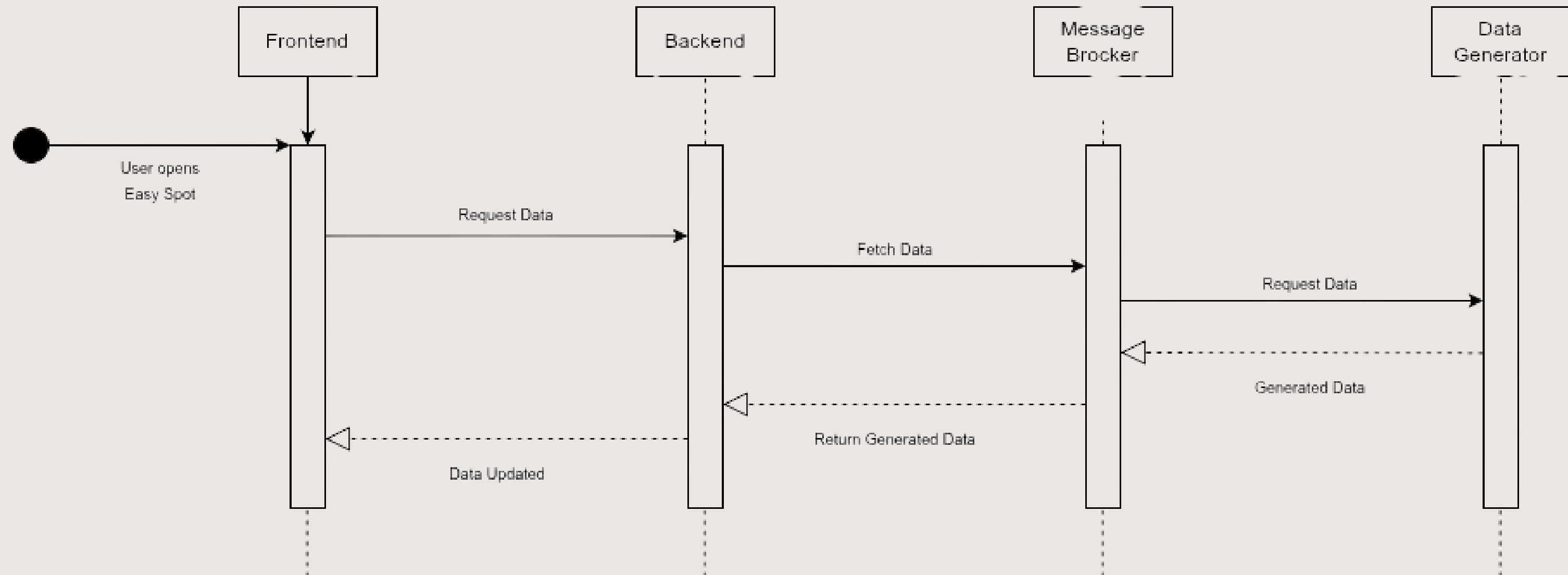
The user interacts with a web application built using React. The app communicates with the backend through REST APIs for actions like searching and booking, and uses WebSockets to receive real-time updates. This allows users to view available parking spots and live changes instantly.

Architecture

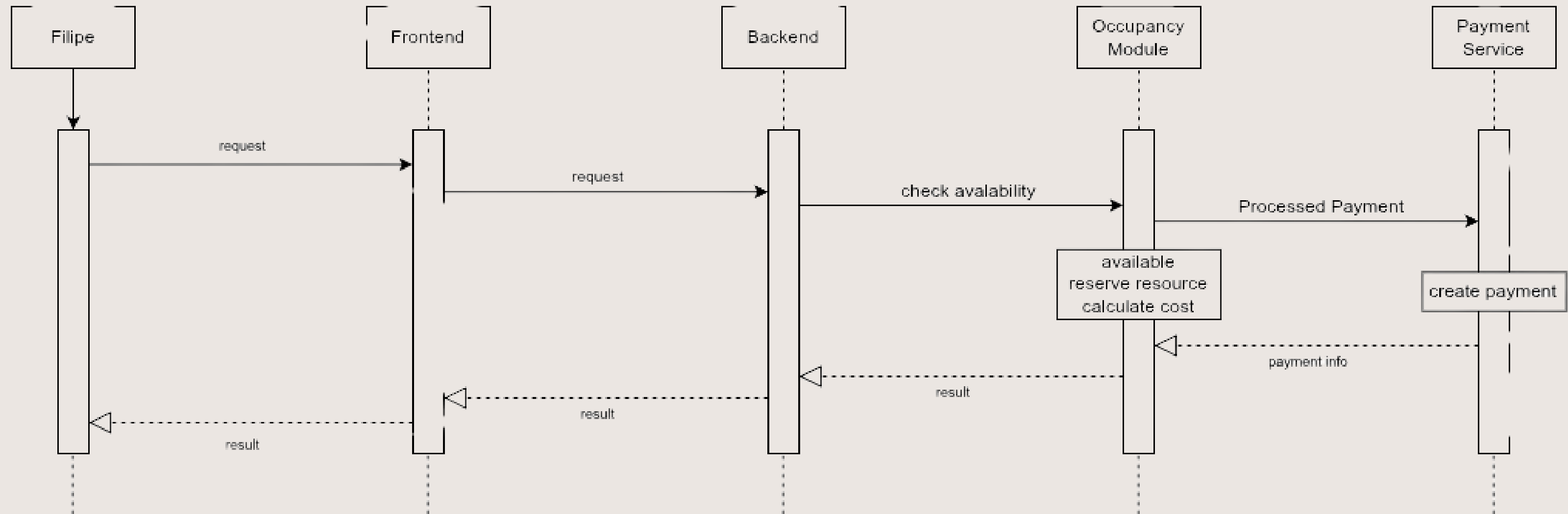
Our architecture is divided into independent modules within the logic layer. Each module communicates with its respective repository, but we also allow for inter-module communication to handle complex business workflows.



User Open Easy Spot



US#3 Spot Reservation



Prototypes

Prototype: Driver

User Story #1 - #11

EasySpot

Tempo Real

Lista de Parques

Mapa

Custos

Favoritos

Perfil

Estacionamento

Pesquisar parque ou zona...

Todos os lugares

Com Lugares

Carregador EV

Acessível

12 parques encontrados

DISPONÍVEL

Estádio Cidade de Coimbra

Rua Dom Manuel I, Coimbra

38% LIVRES 46/120

9 min

€1.80

/HORA

DISPONÍVEL

CoimbraShopping

Av. Mendes Silva, Coimbra

28% LIVRES 17/60

30 min

€1.50

/HORA

DISPONÍVEL

Fórum Aveiro

R. do Batalhão de Caçadores 10, Aveiro

26% LIVRES 22/85

5 min

€1.50

/HORA

DISPONÍVEL

Glicínias Plaza

R. Eng. Von Haff, Aveiro

33% LIVRES 50/150

22 min

€1.00

/HORA

QUASE CHEIO

Europa

Avenida Marquês de Pombal, Leiria

10% LIVRES 9/90

8 min

€1.60

/HORA

DISPONÍVEL

Estádio Municipal Dr....

Caminho da Ribeira, Leiria

40% LIVRES 120/300

18 min

€0.80

/HORA

DISPONÍVEL

Gaivotas

Av. do Mar, Figueira da Foz

43% LIVRES 30/70

10 min

€1.00

/HORA

DISPONÍVEL

Foz Plaza

Rua dos Condados, Buarcos, Figueira da Foz

31% LIVRES 78/250

28 min

€1.20

/HORA

QUASE CHEIO

Estação Ferroviária de Ovar

Largo da Estação, Ovar

21% LIVRES 10/48

4 min

€0.80

/HORA

DISPONÍVEL

Praia do Furadouro

Av. Central, Furadouro

83% LIVRES 100/120

17 min

€0.40

/HORA

DISPONÍVEL

Mercado Municipal de...

Av. José Augusto de Carvalho, Arganil

32% LIVRES 19/60

3 min

€0.60

/HORA

DISPONÍVEL

Montalto

R. do Montalto, Arganil

94% LIVRES 94/100

32 min

€0.30

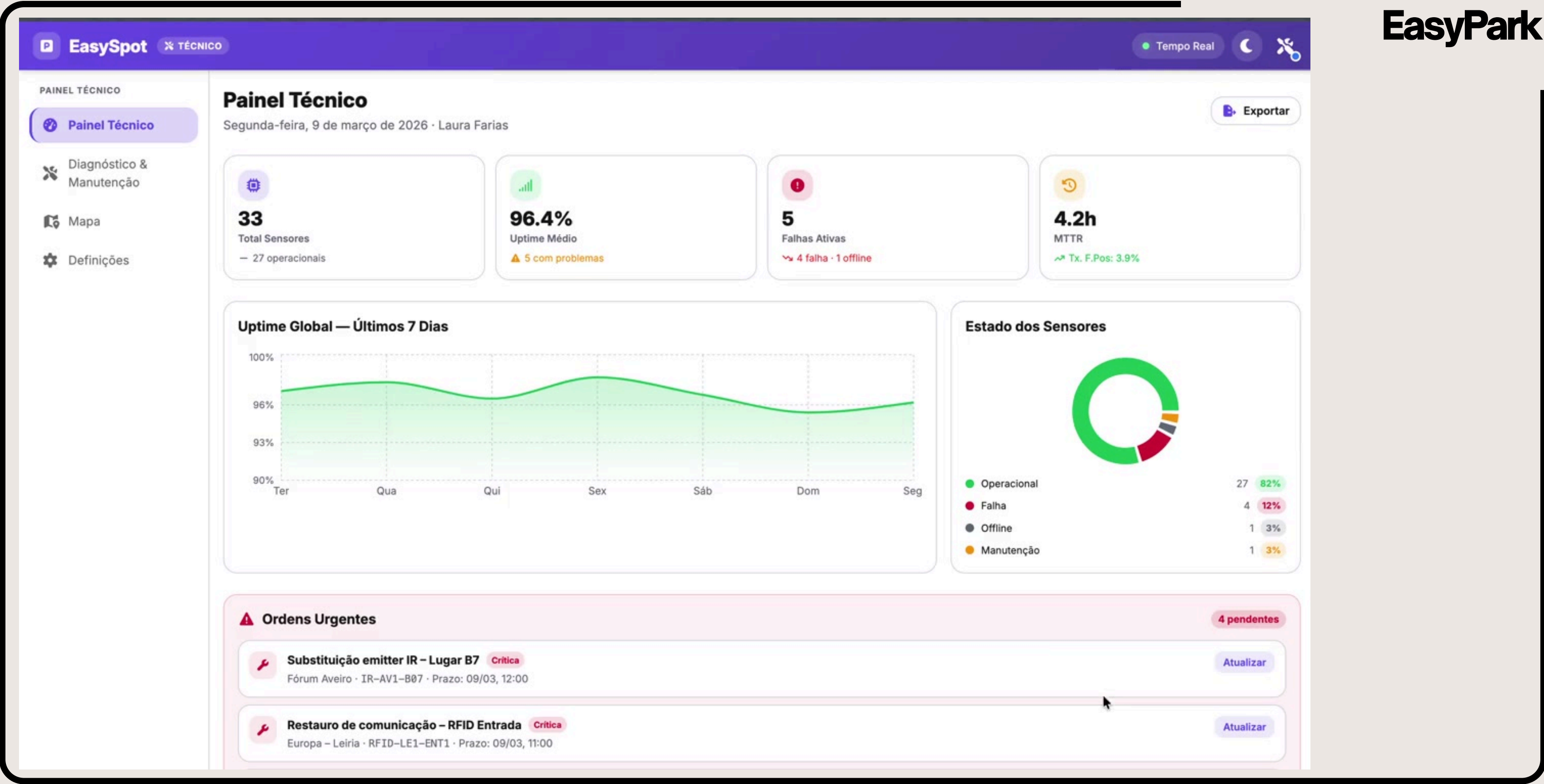
/HORA

http://localhost:5173/parque/coimbra-2

Prototype: Maintenance Technician

User Story #13: Infrastructur Mapping & Status Monitoring

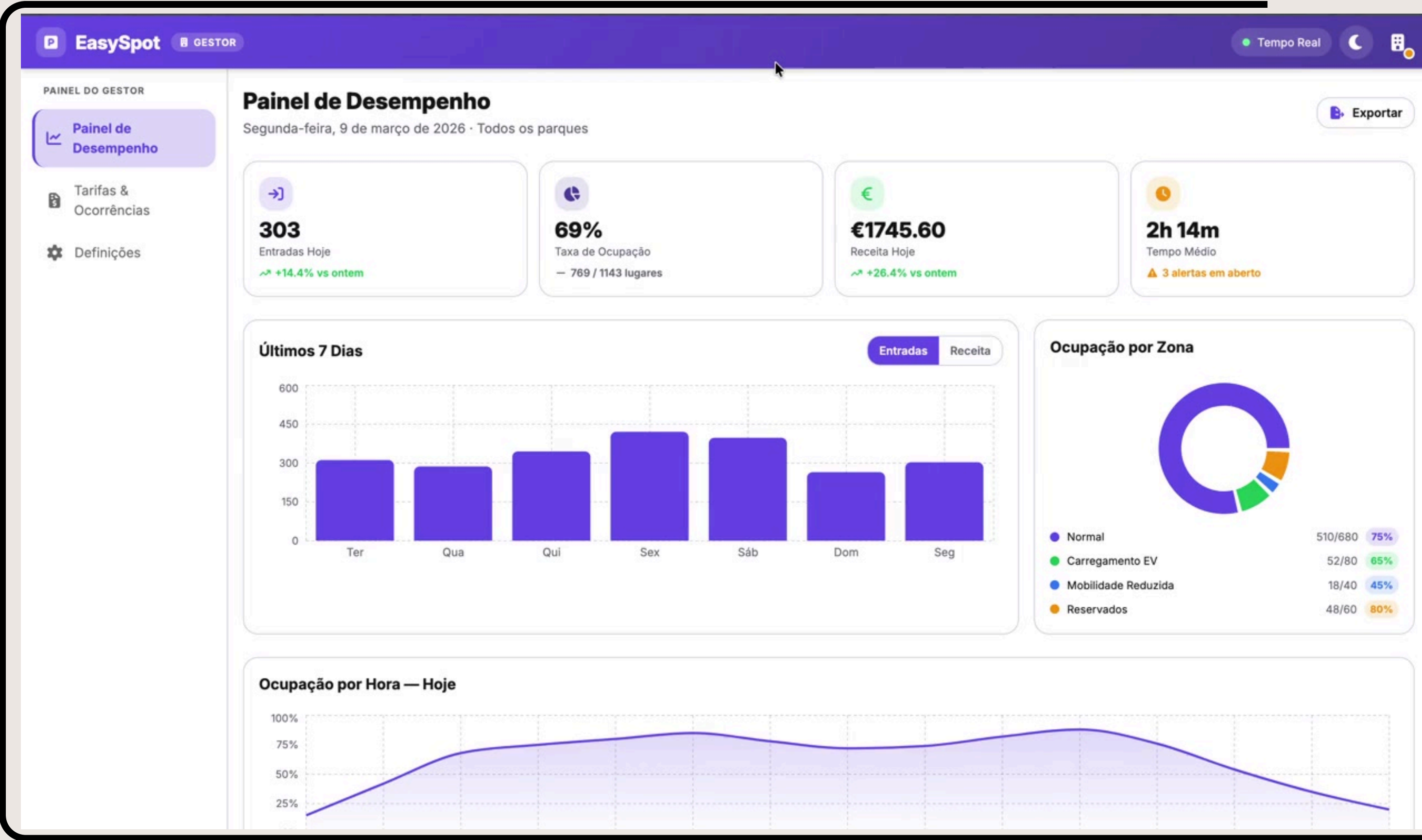
User Story #14: Repair & Update of sensors



Prototype: Parking Operations Manager

User Story #10: Real-Time Occupancy & Financial Performance

User Story #11: Tariff Management & Issue Oversight



thank you

EasyPark

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

